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Display system for remote control of working machine

Abstract

The display system for remote control of a working machine includes a plurality of camera units and an image processing device, both provided on or in a working machine, and a display device and a remote control device, both provided in the remote control side. The display device detects movement of the head of an operator and transmits the detected movement of the head of the operator to the remote control device. The remote control device transmits, to the image processing device over wireless communications, the movement of the head of the operator transmitted from the display device. The image processing device adjusts, for transmission to the display device, parts of left-eye and right-eye panoramic images according to the movement of the head of the operator transmitted from the remote control device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6195204	12/2000	Nalwa	N/A	N/A
6665003	12/2002	Peleg et al.	N/A	N/A
6795109	12/2003	Peleg et al.	N/A	N/A
7218342	12/2006	Kobayashi	348/169	G08B 13/19686
8274550	12/2011	Steuart, III et al.	N/A	N/A
8355041	12/2012	Chen et al.	N/A	N/A
8417059	12/2012	Yamada	348/208.4	H04N 23/68
9715008	12/2016	Cote	N/A	G01S 13/885
9983406	12/2017	Lee	N/A	G02B 27/0093
2001/0038413	12/2000	Peleg et al.	N/A	N/A
2004/0223051	12/2003	Peleg et al.	N/A	N/A
2004/0246333	12/2003	Steuart, III et al.	N/A	N/A
2006/0103723	12/2005	Scire, Jr.	N/A	N/A
2007/0002131	12/2006	Ritchey	N/A	N/A
2007/0014347	12/2006	Prechtl et al.	N/A	N/A
2007/0024701	12/2006	Prechtl et al.	N/A	N/A
2007/0126863	12/2006	Prechtl	348/43	H04N 13/344
2007/0182812	12/2006	Ritchey	N/A	N/A
2008/0024594	12/2007	Ritchey	N/A	N/A
2010/0045773	12/2009	Ritchey	N/A	N/A
2012/0113209	12/2011	Ritchey et al.	N/A	N/A
2012/0327185	12/2011	Steuart III	N/A	N/A
2012/0330849	12/2011	Nielsen	705/301	G06Q 10/20

2013/0063550	12/2012	Ritchey et al.	N/A	N/A
2013/0076599	12/2012	Saito	345/8	G09G 5/00
2014/0118488	12/2013	Steuart, III	N/A	N/A
2014/0364212	12/2013	Osman	463/31	A63F 13/537
2015/0016746	12/2014	Tsubota	382/275	G06T 5/006
2015/0160461	12/2014	Starner	345/156	G02B 27/017
2015/0207991	12/2014	Steuart, III	N/A	N/A
2015/0324692	12/2014	Ritchey et al.	N/A	N/A
2015/0350542	12/2014	Steuart, III	N/A	N/A
2016/0255305	12/2015	Ritchey et al.	N/A	N/A
2017/0061034	12/2016	Ritchey et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000-089395	12/1999	JP	N/A
2001-136518	12/2000	JP	N/A
2003-524927	12/2002	JP	N/A
2003-532914	12/2002	JP	N/A
2004-512701	12/2003	JP	N/A
2005-114235	12/2004	JP	N/A
2005-308282	12/2004	JP	N/A
2007-525054	12/2006	JP	N/A
2008111269	12/2007	JP	N/A
2010-166596	12/2009	JP	N/A
2011-160442	12/2010	JP	N/A
2012-080554	12/2011	JP	N/A
2012-257329	12/2011	JP	N/A
2013-038807	12/2012	JP	N/A
00/39995	12/1999	WO	N/A
01/79908	12/2000	WO	N/A
01/80545	12/2000	WO	N/A
2004/109385	12/2003	WO	N/A
2006/110584	12/2005	WO	N/A
2009/102503	12/2008	WO	N/A
2014/077046	12/2013	WO	N/A

OTHER PUBLICATIONS

International Search Report dated Jul. 14, 2015, issued for PCT/JP2014/002551. cited by applicant
Japanese Office Action dated Jul. 3, 2018 issued in corresponding Japanese Application 2017-522339 cites the patent documents above. cited by applicant

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Background/Summary

.Iadd.CROSS REFERENCES TO RELATED APPLICATIONS .Iaddend.

(1) .Iadd.This application is a reissue application of U.S. Pat. No. 10,474,228, issued on Nov. 12, 2019, which was a national stage application, U.S. Ser. No. 15/524,823, filed on May 5, 2017,

pursuant to 35 U.S.C. § 371 of International Application No. PCT/IB2014/002551, filed on Nov. 14, 2014, the disclosures of which are hereby incorporated by reference in their entirety. .Iaddend.
TECHNICAL FIELD

(2) The present invention relates to a display system that displays an image transmitted from a working machine for remote control of the working machine.

BACKGROUND ART

(3) There have been display systems that display images transmitted from a working machine for remote control of the working machine. The conventional display system includes a camera or a set of cameras that is/are attached to a working machine and controlled remotely from a place located at a distance from a working site where the working machine is at work. The system transmits an image (movie) captured by the camera to a remote control side of the display system over wireless communications so that the display device in the remote control side can display the captured image in real time.

(4) For example, JP2008111269A discloses a display system displaying a panoramic image of the circumstance of a working machine on a display device in real time.

(5) JP2002345058A, as another example, discloses a display system displaying an image of a working site on a display device in real time based on data transmitted from a camera attached to a construction machine. The data transmission is carried out by 2.4 GHz small power radio because of constraints of the Radio Law.

SUMMARY OF INVENTION

Problems to be Solved by the Invention

(6) The display system for remote control of a working machine is hopefully capable of tracking movement of the line of sight of an operator and accordingly changing the image being displayed on a display device to improve remote control operability.

(7) JP2008111269A, however, discloses nothing more than a display system changing panoramic images in response to a pedal operation. JP2008111269A does not always deliver a good operation feeling because the display system does not track movement of the line of sight of the operator to accordingly change the image being displayed on the display device. Neither does JP2002345058A disclose a display system tracking movement of the line of sight of the operator to accordingly change the image being displayed on the display device.

(8) The patent documents listed below represent exemplary technology that is related to the present invention. However, none of them anticipates the present invention.

(9) U.S. Pat. No. 6,795,109B2

(10) U.S. Pat. No. 8,274,550B2

(11) U.S. Pat. No. 6,195,204B1

(12) US2006/0103723A1

(13) US2007/0182812A1

(14) U.S. Pat. No. 8,355,041B2

(15) Accordingly, the present invention has an object to provide a display system for remote control of a working machine, the display system being capable of tracking movement of the line of sight of an operator to accordingly change the image being displayed on a display device.

Solution to Problem

(16) The present invention, to achieve the object, provides a display system for remote control of a working machine, the display system having a first aspect to a fourth aspect.

(17) (1) First Aspect of the Display System

(18) The display system in its first aspect is a display system for remote control of a working machine including: a plurality of camera units attached to a working machine having a working component capable of horizontal rotation and vertical swinging; an image processing device, provided in the working machine, that generates a panoramic image based on a plurality of images captured by the plurality of camera units and transmits a part of the generated panoramic image

over wireless communications; a display device in a remote control side of the display system; and a remote control device that receives the part of the panoramic image from the image processing device and transmits the received part of the panoramic image to the display device, wherein each camera unit includes a pair of a left-eye camera and a right-eye camera, the image processing device generates a left-eye panoramic image and a right-eye panoramic image based respectively on left-eye and right-eye images captured by the pairs of cameras and transmits parts of the generated left-eye and right-eye panoramic images to the remote control device over wireless communications, the remote control device receives the parts of the left-eye and right-eye panoramic images from the image processing device and transmits the received parts of the left-eye and right-eye panoramic images to the display device, the display device, designed to be mounted on the head of an operator, receives the parts of the left-eye and right-eye panoramic images from the remote control device and displays the received parts of the left-eye and right-eye panoramic images, as well as detects movement of eyeballs of the operator and transmits the detected movement of the eyeballs of the operator to the remote control device, the remote control device transmits, to the image processing device over wireless communications, the movement of the eyeballs of the operator transmitted from the display device, and the image processing device adjusts the parts of the left-eye and right-eye panoramic images that are to be transmitted to the display device according to the movement of the eyeballs of the operator transmitted from the remote control device, and if a response speed of the movement of the eyeballs of the operator is faster than a predetermined response speed, the image processing device implements an adjusting process or a cut-out process by delaying the response speed of the movement of the eyeballs of the operator.

(19) (2) Second Aspect of the Display System

(20) The display system in its second aspect is a display system for remote control of a working machine, comprising: a plurality of camera units attached to a working machine having a working component capable of horizontal rotation and vertical swinging; an image processing device, provided in the working machine, that generates a panoramic image based on a plurality of images captured by the plurality of camera units and transmits a part of the generated panoramic image over wireless communications; a display device in a remote control side of the display system; and a remote control device that receives the part of the panoramic image from the image processing device and transmits the received part of the panoramic image to the display device, wherein each camera unit includes a pair of a left-eye camera and a right-eye camera, the image processing device generates a left-eye panoramic image and a right-eye panoramic image based respectively on left-eye and right-eye images captured by the pairs of cameras and transmits parts of the generated left-eye and right-eye panoramic images to the remote control device over wireless communications, the remote control device receives the parts of the left-eye and right-eye panoramic images from the image processing device and transmits the received parts of the left-eye and right-eye panoramic images to the display device, the display device, designed to be mounted on the head of an operator, receives the parts of the left-eye and right-eye panoramic images from the remote control device and displays the received parts of the left-eye and right-eye panoramic images, as well as detects movement of the head of the operator and transmits the detected movement of the head of the operator to the remote control device, the remote control device transmits, to the image processing device over wireless communications, the movement of the head of the operator transmitted from the display device, the image processing device adjusts the parts of the left-eye and right-eye panoramic images that are to be transmitted to the display device according to the movement of the head of the operator transmitted from the remote control device, and the display device displays, along with the parts of the left-eye and right-eye panoramic images, a part of work-assisting information that corresponds to the parts of the left-eye and right-eye panoramic images based on a database prepared in advance in the display system or a database located at a different location than the display system, in such a manner that the part of work-

assisting information coincides with the parts of the left-eye and right-eye panoramic images.

(21) (3) Third Aspect of the Display System

(22) The display system in its third aspect is a display system for remote control of a working machine including: a plurality of camera units attached to a working machine having a working component capable of horizontal rotation and vertical swinging; an image processing device, provided in the working machine, that generates a panoramic image based on a plurality of images captured by the plurality of camera units and transmits the generated panoramic image over wireless communications; a display device in a remote control side of the display system; and a remote control device that receives the panoramic image from the image processing device and transmits a part of the received panoramic image to the display device, wherein each camera unit includes a pair of a left-eye camera and a right-eye camera, the image processing device generates a left-eye panoramic image and a right-eye panoramic image based respectively on left-eye and right-eye images captured by the pairs of cameras and transmits the generated left-eye and right-eye panoramic images to the remote control device over wireless communications, the remote control device receives the left-eye and right-eye panoramic images from the image processing device and transmits parts of the received left-eye and right-eye panoramic images to the display device, the display device, designed to be mounted on the head of an operator, receives the parts of the left-eye and right-eye panoramic images from the remote control device and displays the received parts of the left-eye and right-eye panoramic images, as well as detects movement of eyeballs of the operator and transmits the detected movement of the eyeballs of the operator to the remote control device, the remote control device adjusts the parts of the left-eye and right-eye panoramic images that are to be transmitted to the display device according to the movement of the eyeballs of the operator transmitted from the display device, and if a response speed of the movement of the eyeballs of the operator is faster than a predetermined response speed, the remote control device implements an adjusting process or a cut-out process by delaying the response speed of the movement of the eyeballs of the operator.

(23) (4) Fourth Aspect of the Display System

(24) The display system in its fourth aspect is a display system for remote control of a working machine, comprising: a plurality of camera units attached to a working machine having a working component capable of horizontal rotation and vertical swinging; an image processing device, provided in the working machine, that generates a panoramic image based on a plurality of images captured by the plurality of camera units and transmits the generated panoramic image over wireless communications; a display device in a remote control side of the display system; and a remote control device that receives the panoramic image from the image processing device and transmits a part of the received panoramic image to the display device, wherein each camera unit includes a pair of a left-eye camera and a right-eye camera, the image processing device generates a left-eye panoramic image and a right-eye panoramic image based respectively on left-eye and right-eye images captured by the pairs of cameras and transmits the generated left-eye and right-eye panoramic images to the remote control device over wireless communications, the remote control device receives the left-eye and right-eye panoramic images from the image processing device and transmits parts of the received left-eye and right-eye panoramic images to the display device, the display device, designed to be mounted on the head of an operator, receives the parts of the left-eye and right-eye panoramic images from the remote control device and displays the received parts of the left-eye and right-eye panoramic images, as well as detects movement of the head of the operator and transmits the detected movement of the head of the operator to the remote control device, the remote control device adjusts the parts of the left-eye and right-eye panoramic images that are to be transmitted to the display device according to the movement of the head of the operator transmitted from the display device, and the display device displays, along with the parts of the left-eye and right-eye panoramic images, a part of work-assisting information that corresponds to the parts of the left-eye and right-eye panoramic images based on a database

prepared in advance in the display system or a database located at a different location than the display system, in such a manner that the part of work-assisting information coincides with the parts of the left-eye and right-eye panoramic images.

(25) In an exemplary aspect of the present invention, the plurality of camera units are located at disposition positions that correspond horizontally to a position of a seat that could be installed on the working machine, so that the plurality of camera units have centers of lenses thereof at a seated height that corresponds to a height of the eyes of a working person who, having an average physique for a target market of the working machine, is presumably seated on the seat.

(26) In another exemplary aspect of the present invention, the plurality of camera units are provided on a support member capable of upward/downward movement and vertical rotation.

(27) In another exemplary aspect of the present invention, the display system further includes an illumination device on a periphery of the plurality of camera units.

Advantageous Effects of Invention

(28) The present invention can provide a display system for remote control of a working machine, the display system being capable of tracking movement of the line of sight of an operator to accordingly change the image being displayed on a display device.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic illustration of a display system for remote control of a working machine in accordance with a first embodiment.

(2) FIG. 2 is a schematic perspective view of the structure of a support in a support member provided on the working machine shown in FIG. 1 and a plurality of camera units supported by that support.

(3) FIG. 3 is a block diagram showing a system configuration for an image processing device provided in the working machine side of the display system in accordance with the first embodiment.

(4) FIG. 4 is a diagram showing a cut-out process in which the image processing device cuts out parts of panoramic images.

(5) FIG. 5 is a block diagram showing a system configuration for a remote control device provided in a remote control side of the display system in accordance with the first embodiment.

(6) FIG. 6 is a diagram showing an adjusting process in which the image processing device adjusts parts of panoramic images according to movement of the head of an operator.

(7) FIG. 7 is a flow chart depicting an exemplary control operation implemented by the display system in accordance with the first embodiment.

(8) FIG. 8 is a schematic illustration of a display system for remote control of the working machine in accordance with a second embodiment.

(9) FIG. 9 is a block diagram showing a system configuration for an image processing device provided in the working machine side of a display system for remote control of the working machine in accordance with a third embodiment.

(10) FIG. 10 is a block diagram showing a system configuration for a remote control device provided in the remote control side of the display system in accordance with the third embodiment.

(11) FIG. 11 is a flow chart depicting an exemplary control operation implemented by the display system in accordance with the third embodiment.

(12) FIG. 12 is a schematic illustration of a display system for remote control of the working machine in accordance with a fourth embodiment.

(13) FIG. 13 is a schematic illustration of a display system for remote control of the working machine in accordance with a fifth embodiment.

(14) FIG. 14 is a schematic illustration of a pair of cameras constituting a camera unit and exemplary illumination devices disposed on the pair of cameras for a display system for remote control of the working machine in accordance with a sixth embodiment, as they are viewed from the front.

(15) FIG. 15 is a representation of exemplary display screens on a display device for a display system for remote control of the working machine in accordance with a seventh embodiment, the display screens showing, along with parts of panoramic images, parts of work-assisting information associated with those parts of the panoramic images in such a manner that the parts of work-assisting information coincide with the parts of the panoramic images.

DESCRIPTION OF EMBODIMENTS

(16) The following will describe embodiments in accordance with the present invention in reference to drawings.

First Embodiment

(17) FIG. 1 is a schematic illustration of a display system **100** for remote control of a working machine **200** in accordance with a first embodiment.

(18) As illustrated in FIG. 1, the display system **100** for remote control of the working machine **200** includes camera units **310-310** attached to the working machine **200** and controlled remotely from a place located at a distance from a working site where the working machine **200** is at work. Images (movies) captured by the camera units **310-310** are transmitted to a remote control side **120** of the display system over wireless communications so that the captured images can be displayed on a display device **410** in the remote control side **120** in real time.

(19) In the display system **100**, the working machine **200** is an unmanned working machine controlled remotely from a place located at a distance from a working site by an operator **500** of the working machine **200**. The operator **500**, looking at the display screen of the display device **410**, sends operational instructions from the remote control side **120** to control the unmanned working machine. The working machine **200** is thus enabled to work in dangerous working sites that humans cannot approach (e.g., due to contamination of the sites).

(20) The working machine **200** includes a working component **220** capable of horizontal rotation and vertical swinging. In this example, the working machine **200** is a construction machine (specifically, an excavator). The working machine **200** includes also a traveling body **210** constituting a crawler. The working component **220** is capable of rotating horizontally (around an axis that extends in the vertical direction) relative to the traveling body **210**. The working component **220** includes: an upper body **221** capable of rotating horizontally relative to the traveling body **210**; a boom **222** capable of swinging vertically (around an axis that extends in the horizontal direction) relative to the upper body **221**; an arm **223** capable of swinging vertically relative to the boom **222**; and a bucket **224** capable of swinging vertically relative to the arm **223**. The “horizontal” and “vertical” directions in this context are defined relative to the working machine **200** placed on a horizontal plane.

(21) In this example, the attachment to the end of the arm **223** is the bucket **224**. The example is however by no means limiting: any suitable attachment according to the nature of the work may be fitted to the end of the arm **223**.

(22) In this example, the working machine **200** is an excavator. The example is however by no means limiting: the working machine **200** may be any other construction machine or any agricultural machine.

(23) The working machine **200** further includes a remote operation device **230**. The remote operation device **230** remotely controls a main body of the working machine **200** and is capable of wireless communications with the main body of the working machine **200**. The remote operation device **230** accepts operations from the operator **500** and transmits operational instructions based on the operations to the main body of the working machine **200** over wireless communications. The remote operation device **230** includes an operation section **231**, for example, a control lever with

which the working component **220** (specifically, the upper body **221**, boom **222**, arm **223**, bucket **224**, or any other component) is operated.

(24) The display system **100** is composed of components **300** in the working machine side **110** of the display system and components **400** in the remote control side **120** of the display system.

(25) The components **300** in the working machine side **110** include a plurality of camera units **310-310** and an image processing device **320**, both being provided on the working machine **200**. The components **400** in the remote control side **120** include a display device **410** and a remote control device **420**.

(26) The components **300** in the working machine side **110** further include a support member **330**. The support member **330** has its lower end supported by the upper body **221** (in this example, fixed to the upper body **221**) provided on the working component **220** of the working machine **200** and its upper end supporting (in this example, maintaining) the camera units **310-310** in a horizontal attitude. The “horizontal attitude” in this context is defined as the attitude of the working machine **200** when the working machine **200** is placed on a horizontal plane.

(27) The support member **330** includes a support **331** and a support base (in this example, a post **332**). The support **331** supports the camera units **310-310**. The support base supports the support **331** on its upper end and is in turn supported by the upper body **221** on its lower end.

(28) FIG. 2 is a schematic perspective view of the structure of the support **331** in the support member **330** provided on the working machine **200** shown in FIG. 1 and the camera units **310-310** supported by that support **331**. FIG. 2 shows some of the (five, in this example) camera units **310-310** disposed on the support **331** being removed from the support **331**; in actuality, all of the (five, in this example) camera units **310-310** are disposed on the support **331**.

(29) As illustrated in FIG. 2, each camera unit **310-310** is composed of a pair of a left-eye camera **311** and a right-eye camera **312**. More specifically, the pair of cameras **311** and **312** are composed of the left-eye camera **311** for use with the left eye of the operator **500** and the right-eye camera **312** for use with the right eye of the operator **500**. The pair of cameras **311** and **312** have centers of lenses thereof being separated by a distance “d” that is equal to the distance between the centers of an average person's pupils. In this example, all the cameras **311** and **312** constituting the camera units **310-310** are compact digital video cameras of the same type.

(30) The support **331** is structured to omnidirectionally or substantially omnidirectionally support the camera units **310-310** so that the images captured by the camera units **310-310** can overlap at a predetermined captured area ratio. Specifically, the support **331** is structured to omnidirectionally or substantially omnidirectionally support the left-eye cameras **311-311** so that the left-eye images captured by the left-eye cameras **311-311** can overlap at a predetermined captured area ratio and to omnidirectionally or substantially omnidirectionally support the right-eye cameras **312-312** so that the right-eye images captured by the right-eye cameras **312-312** can overlap at a predetermined captured area ratio.

(31) More specifically, when there are provided two camera units **310** and **310**, the support **331** may allow arranging the camera units **310** and **310** at opposite positions separated by a predetermined distance; when there are provided three or more camera units **310-310**, the support **331** may allow arranging the camera units **310-310** radially at equally or substantially equally separated positions along an imaginary circle with a radius of a predetermined length.

(32) Specifically, the support **331** is platelike and may be shaped to have a number of sides that matches the number of camera units **310**. For example, when there are provided two camera units **310** and **310**, the support **331** may have an increased length in a direction in which the camera units **310** and **310** are arranged back to back in a plan view; when there are provided three or more camera units **310-310**, the support **331** may have a regular polygonal shape with the same number of sides of an equal length as the number of camera units **310** (specifically, when there are provided three camera units **310-310**, the support **331** may have a regular triangular shape in a plan view; when there are provided four camera units **310-310**, the support **331** may have a regular square

shape in a plan view; when there are provided five camera units **310-310**, the support **331** may have a regular pentagonal shape in a plan view; etc.). Alternatively, the support **331** may be circular in a plan view.

(33) In this example, there are provided five camera units **310-310** (in other words, there are provided a total of ten cameras **311** and **312** constituting the camera units **310-310**). These camera units **310-310** are arranged radially at equally or substantially equally separated positions (the center of each side) along an imaginary circle with a radius of a predetermined length as measured from the center of the support **331** that is pentagonal in a plan view. The camera units **310-310** are fixed using fixing members **331a**.

(34) As detailed above, the support **331** supports the camera units **310-310** arranged at opposite positions or radially at equally or substantially equally separated positions. The arrangement enables the camera units **310-310** as a whole to pan covering a 360° horizontal view. The angle of view of each pair of cameras **311** and **312** constituting the camera units **310-310** may be determined in a suitable manner according to the disposition positions of the camera units **310-310**. In this example, each pair of cameras **311** and **312** has a vertical angle of view of 140° (specifically, 70° above an imaginary horizontal reference plane and another 70° below that plane).

(35) The camera units **310-310** (see FIG. 1) are located at disposition positions that correspond horizontally (specifically, both in the front/back direction and the left/right direction) to a position of a seat that could be installed on the working machine **200** (in this example, the seat on which a working person would be seated during operation if the working machine **200** is manned). In addition, the camera units **310-310** are arranged so as to have centers of lenses thereof at a seated height that corresponds to the height of the eyes of a working person (e.g., a working person from Europe, the U.S., or Asia) who, having an average physique (e.g., height and sitting height) for a target market of the working machine **200** (e.g., specifications for the European, North American, or Asian market), is presumably seated on the seat that could be installed on the working machine **200**.

(36) In this example, in the front of the upper body **221**, the boom **222** is provided at the center of the upper body **221** in terms of the left/right direction for a person facing the traveling direction. The support member **330** is hence erected near the front of the upper body **221** either to the right or left of the center in terms of the left/right direction (to the left in the example in FIG. 1).

(37) The support member **330** is further structured to support the camera units **310-310** so that the camera units **310-310** have centers of lenses thereof at a height that corresponds to the height of the eyes of a working person presumably seated on the seat that could be installed on the working machine **200**.

(38) The image processing device **320**, in this example, is provided in the working component **220** of the working machine **200**. This example is however by no means limiting: the image processing device **320** may be provided in the support member **330**, the traveling body **210**, or any other suitable member.

(39) The image processing device **320** generates panoramic images (panoramic movie data) based on a plurality of images (movie data) captured by the camera units **310-310**, cuts out parts of the generated panoramic images, and transmits the cut-out parts of the panoramic images over wireless communications. The image processing device **320**, in this example, is built around a computer (specifically, a personal computer).

(40) FIG. 3 is a block diagram showing a system configuration for the image processing device **320** in the working machine side **110** of the display system **100** in accordance with the first embodiment.

(41) As illustrated in FIG. 3, the image processing device **320** includes a control section **321** executing various programs, computing, and other processes and a memory section **322** storing various data.

(42) The image processing device **320** has a predetermined communication interface for

communications with the camera units **310-310** and is electrically connected to the camera units **310-310** via a communication interface connecting cable **CB1**. This configuration enables the image processing device **320** to receive a plurality of images (movie data) IM-IM and IM-IM captured by the camera units **310-310**.

(43) The communication interface between the camera units **310-310** and the image processing device **320** is, in this example, a USB (universal serial bus) interface (specifically, USB 3.0 compatible interface). The communication interface between the camera units **310-310** and the image processing device **320** is by no means limited to a USB interface and may be any other communication interface. In addition, the communications between the camera units **310-310** and the image processing device **320** are by no means limited to wired communications and may be short-range wireless communications.

(44) The memory section **322** contains a RAM (random access memory) or like volatile memory and a hard disk device, flash memory, or like non-volatile memory. The volatile memory is used, for example, as a working memory by the control section **321** as required in the execution of various computing and other processes. The non-volatile memory is preloaded (pre-installed) with software, including computer programs, that come from an external resource (e.g., a storage medium or an Internet server).

(45) The image processing device **320** further includes a communications section **323** that performs wireless communications (in this example, short-range wireless communications) with a communications section **423** in the remote control device **420** in the remote control side **120**.

(46) Examples of short-range wireless communications include technology capable of short-range communications (approximately from a few meters to 100 meters): namely, wireless LAN (local area network) communications capable of short-range communications over a range of approximately from a few tens of meters to 100 meters and wireless PAN (personal area network) communications capable of short-range communications over a range of approximately from a few meters to a few tens of meters.

(47) An example of wireless LAN communications is IEEE 802.11 compatible wireless LAN communications. An example of wireless PAN communications is IEEE 802.15 compatible wireless PAN communications.

(48) The communications section **323** transmits/receives data to/from the communications section **423** in the remote control device **420** by short-range wireless communications. In this example, the communications section **323** performs IEEE 802.11 compatible wireless LAN communications.

(49) The communications between the image processing device **320** and the remote control device **420** are by no means limited to short-range wireless communications and may be long-range (more than approximately 100 m) wireless communications.

(50) The image processing device **320** generates left-eye and right-eye panoramic images (panoramic movies) PI based respectively on the left-eye and right-eye images (movies) IM-IM and IM-IM captured by the pairs of cameras **311-311** and **312-312** constituting the camera units **310-310**.

(51) Panoramic images are generated by the following technique as an example. For the left eye, a left-eye panoramic image (panoramic movie) PI is generated by combining frame by frame the left-eye images IM-IM captured by the left-eye cameras **311-311** that are arranged omnidirectionally or substantially omnidirectionally to capture overlapping images IM-IM, so as to synthesize a common image in which the left-eye images IM-IM are overlapping. Likewise, for the right eye, a right-eye panoramic image (panoramic movie) PI is generated by combining frame by frame the right-eye images IM-IM captured by the right-eye cameras **312-312** that are arranged omnidirectionally or substantially omnidirectionally to capture overlapping images IM-IM, so as to synthesize a common image in which the right-eye images IM-IM are overlapping. Any publicly known, conventional technique may be used to generate panoramic images PI and PI; for this reason, no further details are given here.

(52) The image processing device **320** stores the generated left-eye and right-eye panoramic images PI and PI temporarily in the memory section **322**.

(53) The image processing device **320** implements a cut-out process to cut out a part PIP of the left-eye panoramic image PI stored in the memory section **322** and a part PIP of the right-eye panoramic image PI stored in the memory section **322**.

(54) FIG. **4** is a diagram showing the cut-out process in which the image processing device **320** cuts out parts PIP and PIP from panoramic images PI and PI. The left-eye and right-eye panoramic images PI and PI have practically the same configuration. FIG. **4**, not distinguishing between the left-eye and right-eye panoramic images PI and PI, illustrates the cut-out process collectively for both left-eye and right-eye images.

(55) It would be easier to understand the cut-out process in which the image processing device **320** cuts out parts PIP and PIP from panoramic images PI and PI if the head **510** of the operator **500** is located at the center of a hollow sphere on the inside of which the panoramic images PI and PI are arranged omnidirectionally or substantially omnidirectionally (in this example, substantially omnidirectionally) as illustrated in FIG. **4**.

(56) The image processing device **320** cuts out the parts PIP and PIP from the panoramic images PI and PI with predetermined cut-out angles (β and γ) that correspond to the view angle of the operator **500** by using a predetermined reference point α (e.g., a forward yawing reference angle ϕ_y , rolling reference angle ϕ_r , or pitching reference angle ϕ_p of the working machine **200**) as a reference point. Cut-out angle (β and γ) settings are made and stored in advance in the memory section **322**.

(57) The predetermined reference point α , for example, when one of the camera units **310-310** is located near the front of the working machine **200**, may be the center pixel of the entire pixel region of an image transmitted from that camera unit **310** located near the front of the working machine **200**. In addition, the predetermined cut-out angles (β and γ) may match the angles that are less than or equal to the angle of view of an average person. For example, the horizontal cut-out angle β may be approximately less than or equal to 200° , and the vertical cut-out angle γ may be approximately less than or equal to 125° (specifically, 75° below and 50° above an imaginary horizontal reference plane).

(58) The image processing device **320** (see FIG. **3**) transmits the cut-out part PIP of the left-eye panoramic image (panoramic movie data) PI and the cut-out part PIP of the right-eye panoramic image (panoramic movie data) PI to the remote control device **420** in the remote control side **120** over wireless communications. More specifically, the image processing device **320** compresses the cut-out parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI and transmits them to the remote control device **420** over wireless communications in real time. In this example, the data transmitted over wireless communications is encrypted.

(59) The remote control device **420** in the remote control side **120** receives the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI from the image processing device **320** and transmits the received parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI to the display device **410**. The remote control device **420**, in this example, is a computer (in this example, a tablet computer).

(60) FIG. **5** is a block diagram showing a system configuration for the remote control device **420** in the remote control side **120** of the display system **100** in accordance with the first embodiment.

(61) As illustrated in FIG. **5**, the remote control device **420** includes a control section **421** executing various programs, computing, and other processes and a memory section **422** storing various data.

(62) The remote control device **420** has a predetermined communication interface for communications with the display device **410** and is electrically connected to the display device **410** via a communication interface connecting cable CB2. This configuration enables the remote control device **420** to transmit the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI to the display device **410**.

(63) The communication interface between the remote control device **420** and the display device **410** is, in this example, an HDMI (high-definition multimedia interface). The communication interface between the remote control device **420** and the display device **410** is by no means limited to an HDMI and may be any other communication interface. In addition, the communications between the remote control device **420** and display device **410** are by no means limited to wired communications and may be short-range wireless communications.

(64) The memory section **422** contains a RAM or like volatile memory and a flash memory or like non-volatile memory. The volatile memory is used, for example, as a working memory by the control section **421** as required in the execution of various computing and other processes. The non-volatile memory is preloaded (pre-installed) with software, including computer programs, that come from an external resource (e.g., a storage medium or an Internet server).

(65) The remote control device **420** further includes a communications section **423** that performs wireless communications (in this example, short-range wireless communications) with the communications section **323** in the image processing device **320** in the working machine side **110**.

(66) The wireless communications performed by the communications section **423** in the remote control device **420** are the same as the wireless communications performed by the communications section **323** in the image processing device **320**; for this reason, no further details are given here.

(67) The remote control device **420**, in this example, is a tablet computer and alternatively may be any other mobile terminal device, for example, a notebook computer, a multifunctional mobile phone handset ("smartphone"), a watch-like terminal device, or a wearable device (a terminal device that can be attached to clothes). The remote control device **420** may be provided in the display device **410**. The remote control device **420** is by no means limited to a mobile terminal device and may be a desktop computer. The operator **500** may be either standing or seated while remotely controlling the working machine **200**.

(68) More specifically, the remote control device **420** stores the received parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI temporarily in the memory section **422** and at the same time transmits them to the display device **410**.

(69) The display device **410** in the remote control side **120** receives the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI from the remote control device **420** and at the same time displays them. More specifically, the display device **410** performs a streaming replay in which the display device **410** simultaneously receives and reproduces (displays) the parts PIP and PIP of the left-eye and right-eye panoramic images (panoramic movies) PI and PI transmitted in real time from the remote control device **420**.

(70) This configuration enables the display device **410** to realize a real-time (live), three-dimensional display ("3D image" or "stereoscopic image"). Any publicly known, conventional technique may be used to realize a three-dimensional display on the display device **410**; for this reason, no further details are given here.

(71) The display device **410** (see FIG. 1) is, in this example, mounted on the head **510** of the operator **500** (specifically, in front of the eyes). More specifically, the display device **410** is an HMD (head-mounted display). The display device **410** is by no means limited to a head-mounted display (HMD) and may be a desktop display device (monitor) or a video projector.

(72) The components **400** in the remote control side **120** further include a line-of-sight detecting section **411** (see FIGS. 1 and 5) that detects movement of the line of sight of the operator **500**. In this example, the line-of-sight detecting section **411** is an attitude detecting section that detects the attitude of the head **510** of the operator **500** (see FIG. 1).

(73) The line-of-sight detecting section **411** is mounted on the head **510** of the operator **500** as illustrated in FIG. 1 and is capable of detecting a yawing angle θ_y around a y-axis which extends in the vertical direction with respect to the head **510** of the operator **500**, a rolling angle θ_r around an r-axis which extends in the front/back direction with respect to the head **510** of the operator **500**, and a pitching angle θ_p around a p-axis which extends in the left/right direction with respect to the

head **510** of the operator **500**, using the reference attitude of the head **510** (e.g., the attitude of the head **510** when the operator **500** is looking in the horizontal direction) as a reference. Reference values may be specified for the reference attitude prior to the detection of the attitude of the head **510** of the operator **500**.

(74) The line-of-sight detecting section **411**, when it is an attitude detecting section as in the current example, may be, for example, one of various sensors including three-dimensional gyrosensors, three-dimensional gravitational acceleration sensors, rotary encoders, and potentiometers. The line-of-sight detecting section **411** may be publicly known, conventional one; for this reason, no further details are given here.

(75) The line-of-sight detecting section **411** (see FIG. 5) transmits data on the detected movement of the head **510** of the operator **500** (specifically, the yawing angle θ_y , rolling angle θ_r , and pitching angle θ_p with respect to the reference attitude of the head **510**) to the remote control device **420**.

(76) In the present embodiment, the line-of-sight detecting section **411** is provided in the display device **410**. In other words, the display device **410** includes the line-of-sight detecting section **411** to detect movement of the head **510** of the operator **500** (specifically, the yawing angle θ_y , rolling angle θ_r , and pitching angle θ_p with respect to the reference attitude of the head **510**) using the line-of-sight detecting section **411** and transmits data on the movement of the head **510** of the operator **500** detected by the line-of-sight detecting section **411** (specifically, the yawing angle θ_y , rolling angle θ_r and pitching angle θ_p with respect to the reference attitude of the head **510**) to the remote control device **420**.

(77) The transmission of the movement of the head **510** of the operator **500** from the line-of-sight detecting section **411** (in this example, the display device **410**) to the remote control device **420** may be performed by communications similar to the communications between the remote control device **420** and the display device **410**. The transmission of the movement of the head **510** of the operator **500** from the line-of-sight detecting section **411** (in this example, the display device **410**) to the remote control device **420** is by no means necessarily performed by wired communications and may be performed by short-range wireless communications.

(78) The remote control device **420** transmits, to the image processing device **320** in the working machine side **110** over wireless communications, the data on the movement of the head **510** of the operator **500** (specifically, the yawing angle θ_y , rolling angle θ_r , and pitching angle θ_p with respect to the reference attitude of the head **510**) transmitted from the line-of-sight detecting section **411** (in this example, the display device **410**).

(79) The image processing device **320** in the working machine side **110** receives the data on the movement of the head **510** of the operator **500** (specifically, the yawing angle θ_y , rolling angle θ_r , and pitching angle θ_p with respect to the reference attitude of the head **510**) from the remote control device **420** and adjusts the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI that are to be transmitted to the display device **410** according to the received movement of the head **510** of the operator **500** (specifically, the yawing angle θ_y , rolling angle θ_r , and pitching angle θ_p with respect to the reference attitude of the head **510**).

(80) FIG. 6 is a diagram showing an adjusting process in which the image processing device **320** adjusts parts PIP and PIP of panoramic images PI and PI according to movement of the head **510** of the operator **500**. The left-eye and right-eye panoramic images PI and PI have practically the same configuration. FIG. 6, not distinguishing between the left-eye and right-eye panoramic images PI and PI, illustrates the adjusting process collectively for both left-eye and right-eye images.

(81) It would be easier to understand the adjusting process in which the image processing device **320** adjusts parts PIP and PIP of panoramic images PI and PI according to the movement of the head **510** of the operator **500** if the head **510** of the operator **500** is located at the center of a hollow sphere on the inside of which the panoramic images PI and PI are arranged omnidirectionally or substantially omnidirectionally (in this example, substantially omnidirectionally) as illustrated in FIG. 6.

(82) The image processing device **320** calculates, for each panoramic images PI and PI , a relative reference point $\Delta\alpha$ (e.g., a relative yawing reference angle $\phi_y+\theta_y$, relative rolling reference angle $\phi_r+\theta_r$, and relative pitching reference angle $\phi_p+\theta_p$ for a reference point in the front of the working machine **200**) that is displaced from a predetermined reference point α (e.g., a forward yawing reference angle ϕ_y , rolling reference angle ϕ_r , or pitching reference angle ϕ_p of the working machine **200**) by the received movement of the head **510** of the operator **500** (specifically, the yawing angle θ_y , rolling angle θ_r and pitching angle θ_p with respect to the reference attitude of the head **510**).

(83) The image processing device **320** cuts out parts PIP and PIP of the adjusted panoramic images PI and PI . More specifically, the image processing device **320** cuts out parts PIP and PIP from panoramic images PI and PI with cut-out angles (β and γ) that correspond to the view angle of the operator **500** by using the calculated relative reference point $\Delta\alpha$ as a reference.

(84) In this example, the line-of-sight detecting section **411** is an attitude detecting section that detects the attitude of the head **510** of the operator **500** and alternatively may be a line-of-sight detecting section that detects the line of sight (eyeballs' movement) of the operator **500**. In the latter case, if the line-of-sight detecting section responds too rapidly to a change of the line of sight of the operator **500**, the image processing device **320** may implement the adjusting process or the cut-out process by delaying the response of the line-of-sight detecting section.

(85) FIG. 7 is a flow chart depicting an exemplary control operation implemented by the display system **100** in accordance with the first embodiment.

(86) In the control operation implemented by the display system **100** in accordance with the first embodiment, the image processing device **320** in the working machine side **110** first generates panoramic images PI and PI from a plurality of images $IM-IM$ and $IM-IM$ captured by the camera units **310-310** as illustrated in FIG. 7 (step **S101**).

(87) Next, the image processing device **320** implements a cut-out process to cut out parts PIP and PIP of the generated panoramic images PI and PI (step **S102**).

(88) Next, the image processing device **320** transmits the cut-out parts PIP and PIP of the panoramic images PI and PI to the remote control device **420** in the remote control side **120** over wireless communications (step **S103**).

(89) Next, the remote control device **420** in the remote control side **120** receives the parts PIP and PIP of the panoramic images PI and PI from the image processing device **320** in the working machine side **110** and simultaneously transmits those parts to the display device **410** (step **S104**).

(90) Next, the display device **410** simultaneously receives and displays the parts PIP and PIP of the panoramic images PI and PI transmitted from the remote control device **420** (step **S105**).

(91) Next, the display device **410** detects the movement of the head **510** of the operator **500** and simultaneously transmits that movement to the remote control device **420** (step **S106**).

(92) Next, the remote control device **420** receives the movement of the head **510** of the operator **500** from the display device **410** and simultaneously transmits that movement to the image processing device **320** in the working machine side **110** over wireless communications (step **S107**).

(93) Next, the image processing device **320** in the working machine side **110** receives the movement of the head **510** of the operator **500** from the remote control device **420** in the remote control side **120** and simultaneously adjusts the parts PIP and PIP of the panoramic images PI and PI according to the movement of the head **510** (step **S108**).

(94) The display system **100** in accordance with the first embodiment then repeats steps **S101** to **S108** above until the process is completed (step **S109**).

(95) In the display system **100** in accordance with the first embodiment described above, the image processing device **320** adjusts the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI that are to be transmitted to the display device **410** according to the movement of the head **510** of the operator **500** transmitted from the remote control device **420**. The display system **100** hence can adjust the three-dimensional view being displayed on the display device **410**.

according to the movement of the head **510** of the operator **500** and can therefore change the image being displayed on the display device **410** by tracking movement of the line of sight of the operator **500**. This configuration can improve remote control operability. Furthermore, the image processing device **320** transmits the parts PIP and PIP of the panoramic images PI and PI to the remote control device **420** over wireless communications. The display system **100** can hence reduce traffic for wireless communications when compared with cases where complete panoramic images PI and PI are transmitted to the remote control device **420** over wireless communications.

(96) A display system in which a single camera itself is moved to display an image on the display device needs time to move the camera when the image being displayed on the display device is to be changed.

(97) In contrast, the display system **100** in accordance with the first embodiment can eliminate the time delay that inevitably occurs in moving the cameras when compared with cases where the display system is configured to move a camera to display an image on the display device because the image processing device **320** stores panoramic images PI and PI (omnidirectional images or substantially omnidirectional images).

(98) In addition, in the first embodiment, the camera units **310-310** are located at disposition positions that correspond horizontally (specifically, both in the front/back direction and the left/right direction) to a position of a seat that could be installed on the working machine **200** and so that the camera units **310-310** have centers of lenses thereof at a seated height that corresponds to the height of the eyes of a working person who, having an average physique (e.g., height and sitting height) for a target market of the working machine **200**, is presumably seated on the seat. The display device **410** can hence display parts PIP and PIP of panoramic images PI and PI that are similar to the real view that the working person seated on the seat would see.

Second Embodiment

(99) FIG. **8** is a schematic illustration of a display system **100** for remote control of the working machine **200** in accordance with a second embodiment.

(100) As illustrated in FIG. **8**, the display system **100** in accordance with the second embodiment differs from the display system **100** in accordance with the first embodiment in that the remote control device **420** in the remote control side **120** is replaced with a plurality of remote control devices **420-420** and that the display device **410** is replaced with a plurality of display devices **410-410** that correspond respectively to the remote control devices **420-420**.

(101) Those members in the display system **100** in accordance with the second embodiment that are practically the same as those in the display system **100** in accordance with the first embodiment are indicated by the same reference numerals. Description will focus on differences from the display system **100** in accordance with the first embodiment.

(102) The image processing device **320** in the working machine side **110** cuts out parts PIP-PIP and PIP-PIP of panoramic images PI and PI and transmits the cut-out parts PIP-PIP and PIP-PIP of the panoramic images PI and PI to the remote control devices **420-420** in the remote control side **120** over wireless communications. The communications between the image processing device **320** in the working machine side **110** and the remote control devices **420-420** in the remote control side **120** are capable of multiplex communications in which a plurality of signals are simultaneously transmitted over a single communications channel. The multiplex communications are by no means limited: examples include FDM (frequency division multiplexing) communications, TDM (time division multiplexing) communications, and other multiplex communications.

(103) The remote control devices **420-420** in the remote control side **120** receive parts PIP-PIP and PIP-PIP of panoramic images PI and PI from the image processing device **320** and transmit the received parts PIP-PIP and PIP-PIP of the panoramic images PI and PI to the associated display devices **410-410**.

(104) The display devices **410-410** in the remote control side **120** receive parts PIP-PIP and PIP-PIP of panoramic images PI and PI from the associated remote control devices **420-420** and display

the received parts PIP-PIP and PIP-PIP of the panoramic images PI and PI. A plurality of line-of-sight detecting sections **411-411** (in this example, a plurality of display devices **410-410**) detect movement of the heads **510-510** of the associated operators **500-500** and transmit the detected movement of the heads **510-510** of the operators **500-500** to the associated remote control devices **420-420**. The remote control devices **420-420** transmit, to the image processing device **320** over wireless communications, the movement of the heads **510-510** of the operators **500-500** transmitted from the associated line-of-sight detecting sections **411-411** (in this example, the display devices **410-410**).

(105) The image processing device **320** in the working machine side **110** adjusts the parts PIP-PIP and PIP-PIP of the panoramic images PI and PI according to the movement of the heads **510-510** of the operators **500-500** transmitted from the remote control devices **420-420**.

(106) The image processing device **320** cuts out the adjusted parts PIP-PIP and PIP-PIP of the panoramic images PI and PI from the panoramic images PI and PI.

(107) The flow of processes implemented by the display system **100** in accordance with the second embodiment is practically the same as the flow of processes implemented by the display system **100** in accordance with the first embodiment shown in FIG. 7, except that the remote control device **420** is replaced with the remote control devices **420-420** and that the display device **410** is replaced with the display devices **410-410**; for this reason, no further details are given here.

(108) In the display system **100** in accordance with the second embodiment, the operators **500-500** can simultaneously view images being displayed on their individual display devices **410-410** in the fields of view that they want. This configuration can improve work safety when compared with cases where a single operator **500** is viewing an image being displayed on his/her display device **410** as in the first embodiment.

Third Embodiment

(109) The display system **100** in accordance with a third embodiment differs from the display system **100** in accordance with the first embodiment in that it is the remote control device **420** in the remote control side **120**, not the image processing device **320** in the working machine side **110**, that implements a cut-out process on panoramic images PI and PI and an adjusting process on parts PIP and PIP of panoramic images PI and PI.

(110) Those members in the display system **100** in accordance with the third embodiment that are practically the same as those in the display system **100** in accordance with the first embodiment are indicated by the same reference numerals. Description will focus on differences from the display system **100** in accordance with the first embodiment.

(111) FIG. 9 is a block diagram showing a system configuration for the image processing device **320** provided in the working machine side **110** of the display system **100** for remote control of the working machine **200** in accordance with the third embodiment. FIG. 10 is a block diagram showing a system configuration for the remote control device **420** provided in the remote control side **120** of the display system **100** in accordance with the third embodiment.

(112) As illustrated in FIG. 9, the image processing device **320** in the working machine side **110** generates and transmits a left-eye panoramic image PI (data on a complete left-eye panoramic image PI) and a right-eye panoramic image PI (data on a complete right-eye panoramic image PI) to the remote control device **420** in the remote control side **120** over wireless communications.

(113) As illustrated in FIG. 10, the remote control device **420** in the remote control side **120** receives the left-eye panoramic image PI (data on a complete left-eye panoramic image PI) and the right-eye panoramic image PI (data on a complete right-eye panoramic image PI) from the image processing device **320**, cuts out a part PIP of the received left-eye panoramic image PI and a part PIP of the received right-eye panoramic image PI, and transmits the cut-out parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI to the display device **410**.

(114) More specifically, the remote control device **420** stores the received left-eye and right-eye panoramic images PI and PI temporarily in the memory section **422**.

(115) The remote control device **420** implements a cut-out process to cut out a part PIP of the left-eye panoramic image PI stored in the memory section **422** and a part PIP of the right-eye panoramic image PI stored in the memory section **422**.

(116) The cut-out process in which the remote control device **420** cuts out PIP and PIP of panoramic images PI and PI is similar to the cut-out process shown in FIG. **4** in which the image processing device **320** cuts out parts PIP and PIP of panoramic images PI and PI; for this reason, no further details are given here.

(117) The remote control device **420** adjusts the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI that are to be transmitted to the display device **410** according to the movement of the head **510** of the operator **500** transmitted from the display device **410**.

(118) The adjusting process in which the remote control device **420** adjusts parts PIP and PIP of panoramic images PI and PI according to the movement of the head **510** of the operator **500** is similar to the adjusting process shown in FIG. **6** in which the image processing device **320** adjusts PIP and PIP of panoramic images PI and PI according to the movement of the head **510** of the operator **500**; for this reason, no further details are given here.

(119) The remote control device **420** cuts out the adjusted parts PIP and PIP of the panoramic images PI and PI from the panoramic images PI and PI.

(120) FIG. **11** is a flow chart depicting an exemplary control operation implemented by the display system **100** in accordance with the third embodiment.

(121) In the control operation implemented by the display system **100** in accordance with the third embodiment, the image processing device **320** in the working machine side **110** first generates panoramic images PI and PI from a plurality of images IM-IM and IM-IM captured by the camera units **310-310** as illustrated in FIG. **11** (step **S201**).

(122) Next, the image processing device **320** transmits the generated panoramic images PI and PI to the remote control device **420** in the remote control side **120** over wireless communications (step **S202**).

(123) Next, the remote control device **420** in the remote control side **120** receives the panoramic images PI and PI from the image processing device **320** in the working machine side **110** and simultaneously implements the cut-out process to cut out parts PIP and PIP of the panoramic images PI and PI (step **S203**).

(124) Next, the remote control device **420** transmits the cut-out parts PIP and PIP of the panoramic images PI and PI to the display device **410** (step **S204**).

(125) Next, the display device **410** simultaneously receives and displays the parts PIP and PIP of the panoramic images PI and PI transmitted from the remote control device **420** (step **S205**).

(126) Next, the display device **410** detects the movement of the head **510** of the operator **500** and simultaneously transmits that movement to the remote control device **420** (step **S206**).

(127) Next, the remote control device **420** receives the movement of the head **510** of the operator **500** from the display device **410** and simultaneously adjusts the parts PIP and PIP of the panoramic images PI and PI according to the movement of the head **510** (step **S207**).

(128) The display system **100** in accordance with the third embodiment then repeats steps **S201** to **S207** above until the process is completed (step **S208**).

(129) In the display system **100** in accordance with the third embodiment, the remote control device **420** adjusts the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI that are to be transmitted to the display device **410** according to the movement of the head **510** of the operator **500** transmitted from the display device **410**. The display system **100** can hence adjust the three-dimensional view being displayed on the display device **410** according to the movement of the head **510** of the operator **500** and can therefore change the image being displayed on the display device **410** by tracking movement of the line of sight of the operator **500**. This configuration can improve remote control operability. Furthermore, the remote control device **420** in the remote control side **120** implements the adjusting process to adjust parts PIP and PIP of

panoramic images PI and PI according to the movement of the head 510 of the operator 500. The display system 100 can hence eliminate the time taken to transmit the movement of the head 510 from the remote control device 420 in the remote control side 120 to the image processing device 320 in the working machine side 110 when compared with cases where the image processing device 320 in the working machine side 110 implements the adjusting process.

(130) A display system in which a single camera itself is moved to display an image on the display device needs time to move the camera when the image being displayed on the display device is to be changed.

(131) In contrast, the display system 100 in accordance with the third embodiment can eliminate the time delay that inevitably occurs in moving the cameras when compared with cases where the display system is configured to move a camera to display an image on the display device because the remote control device 420 stores panoramic images PI and PI (omnidirectional images or substantially omnidirectional images).

Fourth Embodiment

(132) FIG. 12 is a schematic illustration of a display system 100 for remote control of the working machine 200 in accordance with a fourth embodiment.

(133) As illustrated in FIG. 12, the display system 100 in accordance with the fourth embodiment differs from the display system 100 in accordance with the third embodiment in that the remote control device 420 in the remote control side 120 is replaced with a plurality of remote control devices 420-420 and that the display device 410 is replaced with a plurality of display devices 410-410 that correspond respectively to the remote control devices 420-420.

(134) Those members in the display system 100 in accordance with the fourth embodiment that are practically the same as those in the display system 100 in accordance with the third embodiment are indicated by the same reference numerals. Description will focus on differences from the display system 100 in accordance with the third embodiment.

(135) The image processing device 320 in the working machine side 110 transmits panoramic images PI and PI (data on complete panoramic images PI and PI) to the remote control devices 420-420 in the remote control side 120 over wireless communications. The communications between the image processing device 320 in the working machine side 110 and the remote control devices 420-420 in the remote control side 120 are similar the second embodiment; for this reason, no further details are given here.

(136) The remote control devices 420-420 in the remote control side 120 receive panoramic images PI and PI from the image processing device 320, cut out parts PIP-PIP and PIP-PIP of the received panoramic images PI and PI, and transmit the cut-out parts PIP-PIP and PIP-PIP of the panoramic images PI and PI to the associated display devices 410-410.

(137) The display devices 410-410 in the remote control side 120 receive parts PIP-PIP and PIP-PIP of panoramic images PI and PI from the associated remote control devices 420-420 and display the received parts PIP-PIP and PIP-PIP of the panoramic images PI and PI. A plurality of line-of-sight detecting sections 411-411 (in this example, a plurality of display devices 410-410) detect movement of the heads 510-510 of the associated operators 500-500 and transmit the detected movement of the heads 510-510 of the operators 500-500 to the associated remote control devices 420-420.

(138) The remote control devices 420-420 in the remote control side 120 adjust parts PIP-PIP and PIP-PIP of panoramic images PI and PI according to the movement of the heads 510-510 of the associated operators 500-500 transmitted from the associated line-of-sight detecting sections 411-411 (in this example, the display devices 410-410).

(139) Then, the remote control devices 420-420 cut out the adjusted parts PIP-PIP and PIP-PIP of the panoramic images PI and PI from the panoramic images PI and PI.

(140) The flow of processes implemented by the display system 100 in accordance with the fourth embodiment is practically the same as the flow of processes implemented by the display system

100 in accordance with the third embodiment shown in FIG. **11**, except that the remote control device **420** is replaced with the remote control devices **420-420** and that the display device **410** is replaced with the display devices **410-410**; for this reason, no further details are given here.

(141) In the display system **100** in accordance with the fourth embodiment, the operators **500-500** can simultaneously view images being displayed on their individual display devices **410-410** in the fields of view that they want. This configuration can improve work safety when compared with cases where a single operator **500** is viewing an image being displayed on his/her display device **410** as in the third embodiment.

Fifth Embodiment

(142) FIG. **13** is a schematic side view of a display system **100** for remote control of the working machine **200** in accordance with a fifth embodiment.

(143) As illustrated in FIG. **13**, the display system **100** in accordance with the fifth embodiment differs from the display systems **100-100** in accordance with the first to fourth embodiments in that the support member **330** provided on the working machine **200** is replaced with a support member **340** capable of upward/downward movement and vertical rotation.

(144) Those members in the display system **100** in accordance with the fifth embodiment that are practically the same as those in the display systems **100-100** in accordance with the first to fourth embodiments are indicated by the same reference numerals, and no further details are given for those members here.

(145) The support member **340** includes a support **331** and a support base (in this example, a support arm **342**). The support **331** supports the camera units **310-310**. The support base supports the support **331** on its upper end and is in turn supported by the upper body **221** on its lower end, to maintain the camera units **310-310** in a horizontal attitude.

(146) The support arm **342**, in this example, may support the support **331** on the upper end so that the support **331** can freely rotate around axis (first fulcrum **342a**) that extend in the left/right direction and is supported by the upper body **221** on the lower end so that the support arm **342** can freely rotate around axis (second fulcrum **342b**) that extend in the left/right direction. More specifically, the support arm **342** may be configured so that no matter how the support arm **342** rotate around the axis (second fulcrum **342b**) that extend in the left/right direction, the second fulcrum **342b** is always located behind the first fulcrum **342a**. There are provided a plurality of support arms **342-342** (in this example, two support arms **342** and **342**). The support arms **342-342** are structured as parallel links that are disposed parallel to each other in the front/back direction to support the support **331** in such a manner that the support **331** can move up/down. The first fulcrums **342a-342a** and the second fulcrums **342b-342b** of the support arms **342-342** are separated by such distances as to hold the camera units **310-310** in a horizontal attitude. In this example, the first fulcrums **342a-342a** and the second fulcrums **342b-342b** of the support arms **342-342** are positioned on the same horizontal line. The first fulcrums **342a-342a** and the second fulcrums **342b-342b** of the support arms **342-342** are separated by equal distances. This configuration enables the support member **340** to hold the camera units **310-310** in a horizontal attitude no matter how the support arms **342-342** rotate around axes (second fulcrums **342b-342b**) that extend in the left/right direction. The “horizontal attitude” in this context, as described earlier, is defined as the attitude of the working machine **200** when the working machine **200** is placed on a horizontal plane.

(147) The support member **340** includes a driver section **343** that drives the support arms **342-342**. The driver section **343** is actuated in response to an operational instruction transmitted from the remote operation device **230** in the remote control side **120** to rotate the support arms **342-342** and move the camera units **310-310** up/down. This configuration enables the operator **500** to move the camera units **310-310** up/down by remotely controlling the rotation of the support arms **342-342**.

(148) In this example, the support member **340** is structured so that the vertical range of motion **R** of the centers of lenses of the camera units **310-310** can cover a seated height **H** that corresponds to

the height of the eyes of a working person who, having an average physique for a target market of the working machine **200** (e.g., height and sitting height), is presumably seated on a seat. The image processing device **320** may transmit the current (vertical) height of the centers of lenses of the camera units **310-310** to the remote control device **420** which may in turn transmit the current height to the display device **410** so that the display device **410** can display the current height while displaying various indicator positions, for example, the maximum and minimum heights MAX and MIN of the centers of lenses of the camera units **310-310** as well as the seated height H. This configuration enables the operator **500** to recognize the current height of the centers of lenses of the camera units **310-310**.

(149) The support member **340** is structured so that when the centers of lenses of the camera units **310-310** are at the seated height H, the camera units **310-310** are located at disposition positions that correspond horizontally (specifically, in the front/back direction) to a position of a seat that could be installed on the working machine **200**.

(150) The support arm **342** may be structured to fixedly support the support **331** on the upper end of the support arm **342** in such a manner that the support **331** can be moved up/down.

(151) In the display system **100** in accordance with the fifth embodiment, the camera units **310-310** are disposed on the support member **340** capable of moving up/down and rotating in the vertical direction. This configuration enables the operator **500** to adjust the height of the centers of lenses of the camera units **310-310**, and hence the height of the field of view of the operator **500**, when necessary. As a result, for example, the operator **500** can grasp a wider field of view on the surroundings of the working machine **200** due to a field of view that resembles a bird's eye view. In addition, if the support member **340** is actuated to decrease the height of the camera units **310-310**, the working machine **200** can readily enter a working site located in a hole or with a low ceiling. Furthermore, when the operator **500** is controlling the working machine **200** with a direct view of the surroundings (by directly checking the surroundings), the camera units **310-310** can be retracted into the working machine **200** by actuating the support member **340** in such a manner that the camera units **310-310** can be retracted into the working machine **200**. In addition, the support arms **342-342** are structured so that the second fulcrums **342b-342b** are always located behind the first fulcrums **342a-342a**. This configuration enables the camera units **310-310** to unfailingly capture images of a part of the working component **220** that is located ahead of the first fulcrums **342a-342a**.

Sixth Embodiment

(152) The display system **100** in accordance with a sixth embodiment differs from the display systems **100-100** in accordance with the first to fifth embodiments in that illumination devices **350-350** and **350-350** are additionally provided on the periphery (neighborhood or close proximity) of the camera units **310-310**.

(153) FIG. **14** is a schematic illustration of a pair of cameras **311** and **312** constituting a camera unit **310** and exemplary illumination devices **350** and **350** disposed on the pair of cameras **311** and **312** for a display system **100** for remote control of the working machine **200** in accordance with the sixth embodiment, as they are viewed from the front. All the illumination devices **350-350** and **350-350** disposed on the camera units **310-310** have the same structure; FIG. **14** shows illumination devices **350** and **350** disposed on one of the camera units **310-310** as an example.

(154) As illustrated in FIG. **14**, the illumination devices **350** and **350** are disposed on at least a part of the periphery (in this example, on the entire periphery) of the lenses **311a** and **312a** of the pair of cameras **311** and **312** in such a manner that the light-emitting side faces the subject side. In this example, the illumination devices **350** and **350** include a plurality (eight, in this example) of light-emitting elements **351-351** and **351-351** (specifically, light-emitting diodes) disposed on the entire periphery of the lenses **311a** and **312a** of the pair of cameras **311** and **312**. The light-emitting elements **351-351** and **351-351** are disposed equally or substantially equally separated from each other around the lenses **311a** and **312a**.

(155) In the display system **100** in accordance with the sixth embodiment, the illumination devices **350-350** and **350-350** are disposed on the periphery of the camera units **310-310**. Therefore, the light emitted by the illumination devices **350-350** and **350-350** can efficiently prevent the camera units **310-310** from capturing a shadow in captured images.

Seventh Embodiment

(156) That the operator **500** understands work plans (e.g., on the approximate length, width, and depth of excavation), the rotational range of the working component **220** (specifically, the rotational range of the upper body **221** and the swing range of the boom **222**, arm **223**, and bucket **224**), or the locations of gas pipes, water pipes, and other underground piping prior to starting a project is extremely important at working sites for the improvement of work-related issues, such as work efficiency and safety.

(157) From such a viewpoint, the display system **100** in accordance with a seventh embodiment differs from the display systems **100-100** in accordance with the first to sixth embodiments in that the former offers AR (augmented reality) by displaying, along with parts PIP and PIP of panoramic images PI and PI, parts Q1 and Q1 of work-assisting information Q and Q (miscellaneous supplementary information, including work plans, the rotational range of the working component **220**, and the locations of gas pipes, water pipes, and other underground piping) that correspond to the parts PIP and PIP of the panoramic images PI and PI on the display device **410** in such a manner that the parts Q1 and Q1 are associated with the parts PIP and PIP of the panoramic images PI and PI. Augmented reality is technology that enhances information on a real environment with visual information related to the real environment, or more specifically, technology that combines a video of a real environment captured by the camera units **310-310** with a video produced by a computer in association with the real environment in real time in such a manner that the video produced by a computer changes with a change in the video captured by the camera units **310-310** in order to produce a synthesized video.

(158) FIG. **15** is a representation of exemplary display screens D and D on the display device **410** for a display system **100** for remote control of the working machine **200** in accordance with the seventh embodiment, the display screens D and D showing, along with parts PIP and PIP of panoramic images PI and PI, parts Q1 and Q1 of work-assisting information Q and Q associated with the parts PIP and PIP of the panoramic images PI and PI in such a manner that the parts Q1 and Q1 of work-assisting information Q and Q coincide with the parts PIP and PIP of the panoramic images PI and PI. FIG. **15** shows, as the work-assisting information Q and Q, exemplary underground piping, such as gas pipes and water pipes (the parts Q1 and Q1 of the work-assisting information Q and Q) based on a map of a piping network under the ground G and G. The part PIP of the left-eye panoramic image PI and the part PIP of the right-eye panoramic image PI have practically the same configuration. FIG. **15**, not distinguish between the parts PIP and PIP of the left-eye and right-eye panoramic images PI and PI, illustrates processes for augmented reality collectively for both parts.

(159) The remote control devices **420** and **420** in accordance with the first and third embodiments and the image processing device **320** in accordance with the first embodiment add, to cut-out parts PIP and PIP of panoramic images PI and PI, parts Q1 and Q1 of work-assisting information Q and Q that correspond to the cut-out parts PIP and PIP of the panoramic images PI and PI based on a database prepared in advance in the display system **100** (e.g., a database stored in advance in the memory section **422** in the remote control device **420** or in the memory section **322** in the image processing device **320**) or a database located at a different location than the display system **100** (e.g., a database downloaded from an Internet server), in such a manner that the parts Q1 and Q1 of work-assisting information Q and Q coincide with the parts PIP and PIP of the panoramic images PI and PI.

(160) Alternatively, the image processing devices **320** and **320** in accordance with the first and third embodiments and the remote control device **420** in accordance with the third embodiment add, to

non-cut panoramic images PI and PI, work-assisting information Q and Q that corresponds to the non-cut panoramic images PI and PI based on a database prepared in advance in the display system **100** (e.g., a database stored in advance in the memory section **422** in the remote control device **420** or in the memory section **322** in the image processing device **320**) or a database located at a different location than the display system **100** (e.g., a database downloaded from an Internet server), in such a manner that the work-assisting information Q and Q coincides with the panoramic images PI and PI. In the cut-out process in which parts PIP and PIP of the panoramic images PI and PI are cut out, the corresponding parts Q1 and Q1 of the work-assisting information Q and Q are also cut out.

(161) The database may be, for example, a database that is found in CAD (computer aided design) models. Panoramic images PI and PI contain, for example, location information (latitude and longitude) for the working machine **200** that is available with the GNNS (global navigation satellite system). The work-assisting information Q and Q is, for example, information constituting a panoramic image that contains location information (latitude and longitude) that corresponds to the location information (latitude and longitude) of the panoramic images PI and PI.

(162) The display device **410** displays, along with parts PIP and PIP of panoramic images PI and PI, parts Q1 and Q1 of work-assisting information Q and Q that correspond to the parts PIP and PIP of the panoramic images PI and PI in such a manner that the parts Q1 and Q1 of the work-assisting information Q and Q coincide with the parts PIP and PIP of the panoramic images PI and PI.

(163) This configuration enables the operator **500** to appreciate the work-assisting information Q and Q in advance. For example, the operator **500** can recognize whether or not the rotational position of the working component **220** (specifically, the rotational position of the upper body **221** and the swing positions of the boom **222**, arm **223**, and bucket **224**) has reached a predetermined reference level stipulated in the work plan (e.g., has exceeded a reference job depth or a predetermined reference working range), has reached the working range in the working site (e.g., would come into contact with the ceiling of the working site), or has reached underground objects in the working site (e.g., would come into contact with gas pipes, water pipes, and other underground piping Q1).

(164) This augmented reality may be achieved using a publicly known, conventional technique; for this reason, no further details are given here.

(165) In the display system **100** in accordance with the seventh embodiment, the display device **410** displays, along with parts PIP and PIP of panoramic images PI and PI, parts Q1 and Q1 of work-assisting information Q and Q that correspond to the parts PIP and PIP of the panoramic images PI and PI based on a database prepared in advance in the display system **100** or a database located at a different location than the display system **100** in such a manner that the parts Q1 and Q1 of work-assisting information coincide with the parts PIP and PIP of the panoramic images PI and PI. This configuration enables the operator **500** to recognize the work-assisting information Q and Q (miscellaneous supplementary information, including work plans, the rotational range of the working component, or the locations of gas pipes, water pipes, and other underground piping). That in turn can improve work-related issues, such as work efficiency and safety.

(166) The present invention is by no means limited to the embodiments and examples described above and may be carried out in other various forms. Therefore, the embodiments are for illustrative purposes only in every respect and should not be subjected to any restrictive interpretations. The scope of the present invention is defined only by the claims and never bound by the specification. Those modifications and variations that may lead to equivalents of claimed elements are all included within the scope of the invention.

INDUSTRIAL APPLICABILITY

(167) The present invention relates to a display system for remote control of a working machine and is applicable, especially, for the purpose of tracking movement of the line of sight of an operator to accordingly change the image being displayed on a display device.

REFERENCE SIGNS LIST

(168) **100** Display System **110** Working Machine Side **120** Remote control side **200** Working Machine **210** Traveling Body **220** Working Component **221** Upper Body **222** Boom **223** Arm **224** Bucket **230** Remote Operation Device **231** Operation Section **300** Components in Working Machine Side **310** Camera Unit **311** Left-eye Camera **311a** Lens **312** Right-eye Camera **312a** Lens **320** Image Processing Device **321** Control Section **322** Memory Section **323** Communications Section **330** Support Member **331** Support **331a** Fixing Member **332** Post **340** Support Member **342** Support Arm **342a** First Fulcrum **342b** Second Fulcrum **343** Driver Section **350** Illumination Device **351** Light-emitting Element **400** Components in Remote control side **410** Display Device **411** Line-of-sight Detecting Section **420** Remote Control Device **421** Control Section **422** Memory Section **423** Communications Section **500** Operator **510** Head CB1 Connecting Cable CB2 Connecting Cable D Display Screen G Ground H Seated Height IM Image MAX Maximum Height MIN Minimum Height PI Panoramic Image PIP Part of Panoramic Image Q Work-assisting Information Q1 Part of Work-assisting Information R Range of Motion d Distance $\Delta\alpha$ Relative Reference Point α Reference Point β Horizontal Cut-out Angle γ Vertical Cut-out Angle θ_p Pitching Angle θ_r Rolling Angle θ_y Yawing Angle ϕ_p Pitching Reference Angle ϕ_r Rolling Reference Angle ϕ_y Yawing Reference Angle

Claims

.[.1. A display system for remote control of a working machine, comprising: a plurality of camera units attached to a working machine having a working component capable of horizontal rotation and vertical swinging; an image processing device, provided in the working machine, configured to generate a panoramic image based on a plurality of images captured by the plurality of camera units and transmit a part of the generated panoramic image via wireless communication; a display device in a remote control side of the display system; and a remote control device configured to receive the part of the panoramic image from the image processing device and transmit the received part of the panoramic image to the display device, wherein: each camera unit includes a pair of a left-eye camera and a right-eye camera, the image processing device is configured to generate a left-eye panoramic image and a right-eye panoramic image based respectively on left-eye and right-eye images captured by the pairs of cameras and transmit altered left-eye and right-eye panoramic images to the remote control device over wireless communications, the remote control device is configured to receive the altered left-eye and right-eye panoramic images from the image processing device and transmit the altered left-eye and right-eye panoramic images to the display device, the display device, designed to be mounted on the head of an operator, is configured to receive the p altered left-eye and right-eye panoramic images from the remote control device and display the altered left-eye and right-eye panoramic images, the display device includes a line-of-sight detecting section configured to detect movement of eyeballs of the operator and transmit the detected movement of the eyeballs of the operator to the remote control device, the remote control device is configured to transmit, to the image processing device via wireless communication, the movement of the eyeballs of the operator transmitted from the display device, the image processing device is configured to adjust the altered left-eye and right-eye panoramic images that are to be transmitted to the display device according to the movement of the eyeballs of the operator transmitted from the remote control device, and if a response speed of the movement of the eyeballs of the operator is faster than a predetermined response speed, the image processing device is configured to implement an adjusting process or a cut-out process by delaying the response of the line-of-sight detecting section..].

.[.2. The display system set forth in claim 1, wherein the display device is configured to display augmented work-assisting information that based on a database includes the locations of gas pipes, water pipes, and other underground piping embedded in the left-eye and right-eye panoramic

images, in such a manner that the image processing device adjusts the augmented work-assisting information to coincide with the parts of the left-eye and right-eye panoramic images..].

.[.3. The display system as set forth in claim 1, wherein the plurality of camera units are located at disposition positions that correspond horizontally to a position of a seat that could be installed on the working machine, so that the plurality of camera units have centers of lenses thereof at a seated height that corresponds to a height of the eyes of a working person who, having an average physique for a target market of the working machine, is presumably seated on the seat..].

.[.4. The display system as set forth in any one of claim 1, wherein the plurality of camera units are provided on a support member capable of upward/downward movement and vertical rotation..].

.[.5. The display system as set forth in claim 1, the display system further comprising an illumination device on a periphery of the plurality of camera units..].

.[.6. A display system for remote control of a working machine, comprising: a plurality of camera units attached to a working machine having a working component capable of horizontal rotation and vertical swinging; an image processing device, provided in the working machine, configured to generate a panoramic image based on a plurality of images captured by the plurality of camera units and transmit the generated panoramic image over wireless communications; a display device in a remote control side of the display system; and a remote control device configured to receive the panoramic image from the image processing device and transmit a part of the received panoramic image to the display device, wherein: each camera unit includes a pair of a left-eye camera and a right-eye camera, the image processing device is configured to generate a left-eye panoramic image and a right-eye panoramic image based respectively on left-eye and right-eye images captured by the pairs of cameras and transmit the generated left-eye and right-eye panoramic images to the remote control device via wireless communication, the remote control device is configured to receive the left-eye and right-eye panoramic images from the image processing device and transmit parts of the received left-eye and right-eye panoramic images to the display device, the display device, designed to be mounted on the head of an operator, configured to receive the parts of the left-eye and right-eye panoramic images from the remote control device and display the received parts of the left-eye and right-eye panoramic images, the display device includes a line-of-sight detecting section configured to detect movement of eyeballs of the operator and transmit the detected movement of the eyeballs of the operator to the remote control device, the remote control device configured to adjust the parts of the left-eye and right-eye panoramic images that are to be transmitted to the display device according to the movement of the eyeballs of the operator transmitted from the display device, and if a response speed of the movement of the eyeballs of the operator is faster than a predetermined response speed, the remote control device is configured to implement an adjusting process or a cut-out process by delaying the response of the line-of-sight detecting section..].

.[.7. The display system as set forth in claim 6, wherein the display device is configured to display, augmented work-assisting information that includes the locations of gas pipes, water pipes, and other underground piping embedded in the left-eye and right-eye panoramic images based on a database, in such a manner that the image processing device adjusts the augmented work-assisting information to coincide with the parts of the left-eye and right-eye panoramic images..].

.[.8. The display system as set forth in claim 2, wherein the plurality of camera units are located at disposition positions that correspond horizontally to a position of a seat that could be installed on the working machine, so that the plurality of camera units have centers of lenses thereof at a seated height that corresponds to a height of the eyes of a working person who, having an average physique for a target market of the working machine, is presumably seated on the seat..].

.[.9. The display system as set forth in any one of claim 2, wherein the plurality of camera units are provided on a support member capable of upward/downward movement and vertical rotation..].

.[.10. The display system as set forth in any one of claim 3, wherein the plurality of camera units are provided on a support member capable of upward/downward movement and vertical rotation..].

.11. The display system as set forth in any one of claim 8, wherein the plurality of camera units are provided on a support member capable of upward/downward movement and vertical rotation..].

.12. The display system as set forth in claim 2, the display system further comprising an illumination device on a periphery of the plurality of camera units..].

.13. The display system as set forth in claim 3, the display system further comprising an illumination device on a periphery of the plurality of camera units..].

.14. The display system as set forth in claim 4, the display system further comprising an illumination device on a periphery of the plurality of camera units..].

.15. The display system as set forth in claim 8, the display system further comprising an illumination device on a periphery of the plurality of camera units..].

.16. The display system as set forth in claim 9, the display system further comprising an illumination device on a periphery of the plurality of camera units..].

.17. The display system as set forth in claim 1, wherein the line-of-sight detecting section configured to detect a yawing angle around an axis which extends in the vertical direction with respect to the head of the operator, a rolling angle around an axis which extends in the front/back direction with respect to the head of the operator, and a pitching angle around an axis which extends in the left/right direction with respect to the head of the operator..].

.18. The display system as set forth in claim 17, wherein the altered left-eye and right-eye panoramic images are fragments that have been cut out of the left-eye and right-eye panoramic images..].

.19. The display system as set forth in claim 17, wherein the altered left-eye and right-eye panoramic images are cut out of the left-eye and right-eye panoramic images to correspond to a view angle of the operator by using a predetermined reference point..].

.20. The display system as set forth in claim 17, wherein the image processing device is configured to calculate a relative reference point for each left-eye and right-eye panoramic image that is displaced from a predetermined reference point by the movement detected by the line-of-sight detecting section..].

.Iadd.21. A working machine comprising: a camera unit; and an image processing device for executing a cut-out process of an image captured by the camera unit by cutting at a predetermined cut-out angle corresponding to a view angle of an operator, and according to movement of the head or the eyeball of the operator received from an external device, performing an adjusting process to adjust the cut-out angle of the image captured, wherein the image processing device is characterized in that when a response of line-of-sight detection of the movement of the head or eyeball of the operator is faster than a predetermined response speed, the response of the line-of-sight detection is delayed to implement the adjusting process or the cut-out process. .Iaddend.

.Iadd.22. The working machine according to claim 21, further comprising: a support member provided to support the camera unit with a lens of the camera unit positioned at a height corresponding to a line-of-sight height when the operator of the working machine sits on a seat. .Iaddend.

.Iadd.23. A working machine comprising: a camera unit; an image processing device for executing a cut-out process of an image captured by the camera unit by cutting at a predetermined cut-out angle corresponding to a view angle of an operator, and according to movement of the head or the eyeball of the operator received from a display device, when a response of line-of-sight detection of the movement of the head or the eyeball of the operator is faster than a predetermined response speed, the response of the line-of-sight detection is delayed to implement an adjusting process to adjust the cut-out angle of the image captured; and a support member provided to support the camera unit with a lens of the camera unit positioned at a height corresponding to a line-of-sight height when the operator of the working machine sits on a seat. .Iaddend.

.Iadd.24. The working machine according to claim 23, wherein the support member is capable of upward and downward movement and rotation in a vertical direction, provided to support the

camera unit. .Iaddend.

.Iadd.25. The working machine according to claim 23, further comprising: an illumination device provided around the camera unit. .Iaddend.

.Iadd.26. A display system for remote control of a working machine, comprising: a camera unit installed on the working machine; an image processing device that performs a cut-out process that cuts a captured image taken by the camera unit at a predetermined cut-out angle corresponding to a view angle of an operator; and a display device that displays the captured image transmitted from the image processing device, detects movement of the head or the eyeball of the operator and transmits the movement to the image processing device, wherein the image processing device performs an adjusting process for adjusting the cut-out angle of a part of the captured image transmitted to the display device according to the movement of the head or the eyeball of the operator sent from the display device, and when a response of the line-of-sight detection of the movement of the head or the eyeball of the operator is faster than a predetermined response speed, the response of the line-of-sight detection is delayed to implement the adjusting process or the cut-out process. .Iaddend.

.Iadd.27. A display system for remote control of working machine, comprising: a camera unit installed on the working machine; an image processing device that performs a cut-out process that cuts a captured image taken by the camera unit at a predetermined cut-out angle corresponding to a view angle of an operator; and a display device that displays the captured image transmitted from the image processing device, detects movement of the head or the eyeball of the operator, and transmits the movement to the image processing device, and when a response of line-of-sight detection of the movement of the head or the eyeball of the operator is faster than a predetermined response speed, the image processing device delays the response of the line-of-sight detection to implement an adjusting process or a cut-out process; wherein the image processing device adjusts the cut-out angle of the captured image to be transmitted to the display device according to the movement of the head or the eyeball of the operator sent from the display device, the display device additionally displays the work-assisting information corresponding to the captured image together with the captured image in correspondence with the captured image, and the work-assisting information is a work plan or a working range of the working machine. .Iaddend.

.Iadd.28. A display system for remote control of a working machine, comprising: a camera unit installed on the working machine; an image processing device that performs a cut-out process that cuts a captured image taken by the camera unit at a predetermined cut-out angle corresponding to a view angle of an operator; and a display device that displays the captured image transmitted from the image processing device, detects movement of the head or the eyeball of the operator, and transmits the movement to the image processing device, wherein the image processing device adjusts the cut-out angle of the captured image transmitted to the display device according to the movement of the head or the eyeball of the operator sent from the display device, and when a response of line-of-sight detection of the movement of the head or the eyeball of the operator is faster than a predetermined response speed, the response of the line-of-sight detection is delayed to implement an adjusting process or a cut-out process; and a support member provided to support the camera unit with a lens of the camera unit provided at a height position corresponding to a line-of-sight height when the operator of the working machine sits on a seat. .Iaddend.

.Iadd.29. The display system for remote control of a working machine according to claim 26, further comprising: a support member capable of upward and downward movement and rotation in a vertical direction, provided to support the camera unit. .Iaddend.

.Iadd.30. The display system for remote control of a working machine according to claim 26, further comprising: an illumination device provided around the camera unit. .Iaddend.
